一些常见结论

Contents

1 Abstract Algebra 抽象代数

Proposition 1.1. *Let S be a monoid. If an element $a \in S$ satisfies that there exists $b, c \in S$ s.t. $ba = 1, ac = 1$, then $b = c$.*

Proposition 1.2. *Let R be an integral domain. Then $R = \cap_\alpha R_{\mathfrak{m}_\alpha}$, where $\mathfrak{m}_\alpha$ varies over all the maximal ideals of R.*

Theorem 1.1 (***Schur' Theorem***)**.** *Let G be a group. If $Z(G)$ is of finite index in G, then $G' = [G, G]$ is finite.*

Proposition 1.3. *G is a infinite group. If every nontrivial proper subgroup of G is of finite index, then G is cyclic.*

Proposition 1.4. *G is a finite group, $H \leq G$ subgroup. Show there is a complete set of left cosets being a complete set of right cosets at the same time.*

Proposition 1.5. *All groups of order pq are either isomorphic to $\mathbb{Z}_{pq}$ (abelian case) or $\mathbb{Z}_p \rtimes \mathbb{Z}_q$ (nonabelian case).*

Proposition 1.6. *G is a p-group. Then all maximal subgroups of G is normal and of index p.*

Proposition 1.7. *G is a finite group. Assume that $v_p(G) = n$. Then the number of subgroups of order p^i, n_1, satisfies that $n_1 \equiv 1 \pmod p$, for $\forall 1 \leq i \leq n$.*

Proposition 1.8. *G is a nonabelian simple group of order 60. Then $G \cong A_5$.*

Proposition 1.9. *G is a finite group, $N \trianglerighteq G$ a normal group, $p \mid \sharp(N)$. Assume that P is a Sylow p-group of N. Then $G = N_G(P)N$.*

2 Topology 拓扑

2.1 Basic topology

Proposition 2.1. *Let X be a topological space., $S \subseteq X$ subset. Then S is closed in X if and only if X can be covered by open subsets U_α s.t. $S \cap U_\alpha$ is closed in U_α.*

2.2 Irreducibility

Proposition 2.2. *Let X be an irreducible topological space. Then for all nonempty open subset $U \subseteq X$, U is dense.*

Proposition 2.3. *Let X be a topological space. Then every maximal irreducible subspace of X is closed.*

Remark 2.1. *For convenience, we call maximal irreducible subspace irreducible component.*

Proposition 2.4. *Let X be a noetherian topological space. Then X has only finitely many irreducible components.*

2.3 Dimension theory

3 Commutative Algebra 交换代数

In this section, all rings are commutative.

3.1 Prime ideals

⋆ Radical

Proposition 3.1. *Let R be a ring. The nilradical $\mathfrak{N}$ is the intersection of all prime ideals of R.*

Proposition 3.2. *Let R be a ring. Define the Jacobson radical $\mathfrak{R}$ is the intersection of all maximal ideals of R. We have $x \in \mathfrak{R} \iff 1 - xy$ is a unit in R for all $y \in R$.*

Proposition 3.3. *Let R be a ring, $I, J \subseteq R$ ideals. If $\sqrt{I}$ and $\sqrt{J}$ are coprime, then I and J are coprime.*

Proposition 3.4. *Let R be a ring. Then every idempotent of $R/\mathfrak{N}$ lifts to some idempotent of R. In fact, the lifting is unique.*

⋆ Associated Prime Ideals

Proposition 3.5. *Let R be a noetherian ring, $\mathfrak{p}$ a minimal prime ideal of R. Then there is an element $a \in R$ such that $\operatorname{ann}(a) = \mathfrak{p}$ i.e. minimal prime ideals are associated prime ideals. In particular, we can take $a \notin \mathfrak{p}$.*

Proposition 3.6 (***Structure Theorem for Artinian Rings***)**.** *Let R be an artinian ring. Assume $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ are all prime ideals of R. Then $R \cong \prod_{i=1}^{n} R_{\mathfrak{p}_i}$.*

3.2 Localization

Proposition 3.7. *Let R be a ring. Then $R = \cap_{\mathfrak{m} \in Max(R)} R_{\mathfrak{m}}$.*

Proposition 3.8. *Let R be a reduced ring. Then $S^{-1}R$ is reduced for all multiplicatively closed subset S.*

Corollary 3.1. *Let R be a reduced ring. Then $R_{\mathfrak{p}}$ is a field for all minimal prime ideal $\mathfrak{p}$.*

3.3 Modules

⋆ Finite Generatedness

Proposition 3.9. *Let R be a noetherian ring. Then submodule of finitely generated R-module is still finitely generated.*

Proposition 3.10. *Let R be a noetherian ring. Then for every finite R-module M, there is a filtration*

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M \tag{1}$$

such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$.

Corollary 3.2. *Let R be a noetherian ring. There is a filtration*

$$0 = I_0 \subset I_1 \subset \cdots \subset I_n = R \tag{2}$$

such that $I_i/I_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$. And for all minimal prime ideal $\mathfrak{q}$ of R, $\mathfrak{q}$ appears exactly length $A_{\mathfrak{q}}$ *times.*

⋆ **Flatness**

Proposition 3.11. *Let R be a ring, $M \in \mathbf{Mod}_R$. Then the following conditions are equivalent*
(1)M is finitely presented and flat
(2)M is finite projective
(3)M is locally free of finite rank.

Remark 3.1. *For more general statement, refer to Stacks Project 00NX.*

3.4 Algebras

⋆ **Finite Generatedness**

⋆ **Integrality**

Proposition 3.12. *Let L/K be an algebraic field extension, $A \subseteq B$ and $\operatorname{Frac} A = K$. Assume B is the integral closure of A in L. Then $\operatorname{Frac} B = L$.*

Proposition 3.13. *Let L/K be an algebraic field extension, $A \subseteq B$ and $\operatorname{Frac} A = K$. Assume B is the integral closure of A in L. Then $B \otimes_A K = L$.*

Proposition 3.14. *Let $A \subsetneq B$ be an integral extension of integral domains. Then A is a field if and only if B is a field.*

Corollary 3.3. *Let $A \subseteq B$ be an integral extension. Assume $\mathfrak{P}_1, \mathfrak{P}_2$ are different prime ideals of B, both lying over $\mathfrak{p} \in \operatorname{Spec} A$. Then neither $\mathfrak{P}_1 \subset \mathfrak{P}_2$ nor $\mathfrak{P}_1 \supset \mathfrak{P}_2$.*

Corollary 3.4. *Let A be a ring integral over k. Then $\dim A = 0$. Moreover, if A is noetherian, then A is artinian.*

Corollary 3.5. *Let A be a ring finite over k. Then A is artinian.*

Proposition 3.15. *Let A be a reduced noetherian ring, B integral over A. Assume that for every minimal prime ideal $\mathfrak{p}$ of A, there is some minimal prime ideal $\mathfrak{q}$ of B lying over $\mathfrak{p}$ such that the natural map $k(\mathfrak{p}) \hookrightarrow k(\mathfrak{q})$ is isomorphic. Then the normalization map $A \hookrightarrow \widetilde{A}$ factors through $A \to B$.*

⋆ **Finiteness**

Proposition 3.16. *Let B be a finite A-algebra with homomorphism $\varphi : A \to B$. Assume $\mathfrak{m}$ is a maximal ideal in A. Then $B_\mathfrak{m} \cong \oplus_i B_{\mathfrak{p}_i}$, where $\mathfrak{p}_i$ varies over all prime ideals of B lying over $\mathfrak{m}$.*

⋆ **Flatness**

Proposition 3.17. *Let $f : A \to B$ be a ring homomorphism. Then f is flat if and only if for all $\mathfrak{P} \in \operatorname{Spec} B$ lying over some $\mathfrak{p} \in \operatorname{Spec} A$, the induced map $A_\mathfrak{p} \to B_\mathfrak{P}$ is flat.*

Proposition 3.18. *Let $f : A \to B$ be an etale homomorphism i.e. flat amd unramified. If A is regular, then B is regular.*

Proposition 3.19. *Let $f : A \to B$ be a homomorphism between local rings. Then f is flat if and only if f is faithfully flat.*

⋆ **Faithful Flatness**

Proposition 3.20. *Let $f : A \to B$ be a faithfully flat homomorphism, $M \in \mathbf{Mod}_A$. Then $M = 0$ if and only if $M \otimes_A B = 0$.*

Corollary 3.6. *Let $f : A \to B$ be a faithfully flat homomorphism. Then for each ideal $I \subseteq A$, we have $IB \cap A = I$.*

Corollary 3.7 (***Faithfully Flat Descent***)**.** *Let $f : A \to B$ be a faithfully flat homomorphism. Then for all homomorphism $\varphi : M \to N$ between A-modules, φ is an isomorphism if and only if $\varphi \otimes_A B$ is an isomorphism.*

3.5 Dimension theory

Proposition 3.21. *Let R be an artinian integral domain. Then R is a field.*

Proposition 3.22. *Let A be an irreducible algebra of finite type over field k. Then for any field extension K/k, $A \otimes_k K$ is of pure dimension $\dim A$.*

Corollary 3.8. *Let A be an algebra of finite type over field k. Then for any field extension K/k, $\dim A \otimes_k K = \dim A$.*

Corollary 3.9. *Let A be a k-algebra, K/k finite extension. Then $\dim(A \otimes_k K) = \dim A$.*

3.6 Length

3.7 Local rings

Proposition 3.23. *Let R be a local ring, $\mathfrak{m} \subseteq R$ the maximal ideal. If $\dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = n$, then $\mathfrak{m}$ is generated by exactly n elements.*

Corollary 3.10. *Let R be a local ring, $\mathfrak{m} \subseteq R$ the maximal ideal. If $\dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$, then $\mathfrak{m}$ is principal.*

Proposition 3.24. *Let $(R, \mathfrak{m})$ be a local ring with residue field k, M finitely generated R-module. Then the following conditions are equivalent:*
(1)M is free
(2)M is projective
(3)M is flat
(4)$\mathrm{Tor}_1^R(M, k) = 0$.

Corollary 3.11. *Let R be a DVR, M finite over R. Then M is flat over R if and only if it is torsion-free over R.*

$\star$ **Valuation Rings**

Proposition 3.25. *Let R be a valuation ring, $a, b \in R$. Then the ultrametric inequality $v(a+b) \geq \min\{v(a), v(b)\}$ takes equality whenever $v(a) \neq v(b)$.*

Proposition 3.26. *Let A be a noetherian ring, $\mathfrak{p} \subseteq \mathfrak{P}$ prime ideals of A. Then there is a ring homomorphism $A \to R$ for some DVR R such that under the corresponding spectrum map, (0) is sent to $\mathfrak{p}$ and the maximal ideal $\mathfrak{m}$ is sent to $\mathfrak{P}$.*

$\star$ **Artinian Local Rings**

Proposition 3.27. *Let R be an artinian local ring over k, $\mathfrak{m}$ maximal ideal. Then $\dim_k R = \mathrm{length}(R) \cdot [R/\mathfrak{m} : k]$.*

Proposition 3.28. *Let R be an artinian local ring of finite type over k, $\mathfrak{m}$ maximal ideal, K/k field extension. Then $R \otimes_k K \cong \prod_i R_i$ for R_i artinian local rings with maximal ideals $\mathfrak{m}_i$, and if moreover R is finite over k, we have that*

$$\mathrm{length}(R) \cdot [R/\mathfrak{m} : k] = \sum_i \mathrm{length}(R_i) \cdot [R_i/\mathfrak{m}_i : K] \tag{3}$$

In particular, if $R = L$ is a field, we get $[L : k] = \sum_i \mathrm{length}(R_i) \cdot [R_i/\mathfrak{m}_i : K]$.

Proposition 3.29. *Let R be an artinian local ring over k, $\mathfrak{m}$ maximal ideal, K/k finite extension. Then $R \otimes_k K \cong \prod_i R_i$ for R_i artinian local rings with maximal ideals $\mathfrak{m}_i$, and if moreover R is finite over k, we have that*

$$\mathrm{length}(R) \cdot [R/\mathfrak{m} : k] = \sum_i \mathrm{length}(R_i) \cdot [R_i/\mathfrak{m}_i : K] \tag{4}$$

In particular, if $R = L$ is a field, we get $[L : k] = \sum_i \mathrm{length}(R_i) \cdot [R_i/\mathfrak{m}_i : K]$.

$\star$ **Henselian Local Rings**

Proposition 3.30. *Let R be a local ring. Then R is Henselian if and only if any finite R-algebra is product of local rings.*

Corollary 3.12. *Let R be a Henselian local ring, S finite R-algebra. If S is an integral domain, then S is a local ring.*

Proposition 3.31. *Let R be UFD. Then R is integrally closed.*

Proposition 3.32. *Let $B \in Alg_A$ with ring homomorphism $f : A \longrightarrow B$. For $\mathfrak{p} \in \operatorname{Spec} A$, if $\mathfrak{p}^{ec} = \mathfrak{p}$, then $\mathfrak{p} \in im(f^*)$.*

Proposition 3.33. *Let A be a graded ring and M, N, L graded A-modules. Then $\operatorname{Hom}(M \otimes_A N, L) \cong \operatorname{Hom}(M, \operatorname{Hom}(N, L))$.*

Proposition 3.34. *Let $\varphi : A \longrightarrow B$ be a homomorphism between noetherian rings. Then for each minimal prime ideal $\mathfrak{p}$ containing $\ker(\varphi)$ such that $\varphi(\mathfrak{p}B) \neq B$, we can find $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$.*

Corollary 3.13. *Let $\varphi : A \longrightarrow B$ be a homomorphism between noetherian rings. Then for each prime ideal $\mathfrak{p}$ containing $\ker(\varphi)$ such that $\varphi(\mathfrak{p}B) \neq B$, we can find $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$.*

4 Linear Algebra 线性代数

Proposition 4.1. *If U, V are subspaces of dimension r,s of a vector space of W of dimension n, then $U \cap V$ is a subspace of dimension more than r + s - n.*

5 Representation Theory 表示论

Theorem 5.1 (***Bornside** $p^a q^b$ **Theorem***)**.** *G is a finite group of order $p^a q^b$, where p, q are prime numbers. Then G is solvable group.*

Theorem 5.2 (***Clifford Theorem***)**.** *G is a finite group, $N \trianglerighteq G$ a normal subgroup. Assume that $\chi \in Irr(G)$ and $\varphi \in Irr(N)$ satisfy that φ is a direct summand of $Res_N^G(\chi)$. Then $Res_N^G(\chi) = e \sum_{i=1}^n \varphi^{g_i}$, where $e \in \mathbb{N}$, φ^{g_i} is given by $x \longmapsto \varphi(g_i x g_i^{-1})$ and $g_1, \cdots, g_n$ is a transversal of N in G.*

6 Algebraic Geometry 代数几何

6.1 Classic examples

Proposition 6.1. *Let $\mathcal{L}$ be an invertible sheaf on projective line $\mathbb{P}_k^1$. Then $\mathcal{L}$ must be isomorphic to $\mathcal{O}_{\mathbb{P}_k^1}(n)$ for some n.*

6.2 Spectrum

Proposition 6.2. *Let $\varphi : A \to B$ be a ring homomorphism. Assume that $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is the corresponding morphism of spectrum. Then $\operatorname{im} f \subseteq V(\ker \varphi)$.*

Corollary 6.1. *Let $\varphi : A \to B$ be a homomorphism between noetherian rings. Assume that $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is the corresponding morphism of spectrum. Then $\operatorname{im} f = \{\mathfrak{p} \in V(\ker\varphi) \big| \varphi(\mathfrak{p})B \neq B\}$.*

Corollary 6.2. *Let $\varphi : A \to B$ be a ring homomorphism. Assume that the corresponding morphism $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective. Then $\ker\varphi \subseteq \mathfrak{N}_A$.*

Proposition 6.3. *Let $\varphi : A \to B$ be a surjective ring homomorphism. Then the corresponding morphism $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is injective.*

Proposition 6.4. *Let $\varphi : A \to B$ be a ring homomorphism. Assume that $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is the corresponding morphism of spectrum. Then f is a closed immersion if and only if φ is surjective.*

6.3 Sheaves

⋆ Quasi-coherent Sheaves

Proposition 6.5. *Let X be a locally noetherian scheme, $\mathcal{F}$ coherent sheaf on X. Then $\operatorname{Supp}\mathcal{F} := \{x \in X \big| \mathcal{F}_x \neq 0\}$ is a closed subset of X.*

Remark 6.1. *For one single section s, there are two meaning of support of s over X. One is the set $\{x \in X \big| s_x \neq 0\}$. And the other is the set $\{x \in X \big| s(x) = 0\}$. Both of the sets are closed subset of X. But support of a sheaf is not closed in general.*

⋆ Flatness

Proposition 6.6. *Let $f : X \to Y$ be a morphism. Assume $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is an exact sequence of sheaf modules on X, and $\mathcal{H}$ is flat over Y. Then for every $y \in Y$, the restricted sequence*

$$0 \longrightarrow \mathcal{F}\big|_{X_y} \longrightarrow \mathcal{G}\big|_{X_y} \longrightarrow \mathcal{H}\big|_{X_y} \longrightarrow 0 \tag{5}$$

is still exact.

Proposition 6.7. *Let X be an arbitrary scheme, $\mathcal{F}$ sheaf module on X. If $\mathcal{F}$ is of finite presentation and flat, then $\mathcal{F}$ is locally free.*

Corollary 6.3. *Let X be a noetherian scheme, $\mathcal{F}$ coherent sheaf on X. If $\mathcal{F}$ is flat over X, then $\mathcal{F}$ is locally free.*

6.4 Properties of morphisms

⋆ Separatedness

Proposition 6.8. *Let X be a reduced scheme and Y be a separated scheme. Assume f and g are two morphisms from X to Y, agreeing on an open dense subset of X. Then $f = g$.*

Remark 6.2. *With this proposition, the equivalence classes of rational classes from reduced scheme to separated scheme are well behaved. This is why we always assume schemes are reduced and separated when we consider birational maps.*

Proposition 6.9. *Let $f : Y \to X$ be a separated morphism, X_0 open subscheme of X. Assume the immersion $i : X_0 \hookrightarrow X$ factors as $X_0 \xrightarrow{g} Y \xrightarrow{f} X$. Take the schematic image Y_0 of g with closed immersion $j : Y_0 \hookrightarrow Y$. Then $(f \circ j)^{-1}(X_0) \to X_0$ is isomorphic.*

⋆ **Properness**

Proposition 6.10. *Let $f : X \to Y$ be a proper morphism between noetherian schemes, $\mathcal{F}$ coherent sheaf on X. Then $R^i f_* \mathcal{F}$ is still coherent for all $i \geq 0$.*

⋆ **Integrality**

Proposition 6.11. *Let $f : Y \to X$ be a dominant integral morphism between reduced schemes. Assume X is noetherian and that for every generic point x_g of X, there is some generic point y_g of Y such that $k(x_g) \hookrightarrow k(y_g)$ is isomorphic. Then the normalization map $\widetilde{X} \to X$ factors through f.*

Corollary 6.4. *Let $f : Y \to X$ be a dominant integral morphism, X reduced noetherian. Assume that for every generic point x_g of X, there is some generic point y_g of Y such that $k(x_g) \hookrightarrow k(y_g)$ is isomorphic. Then the normalization map $\widetilde{X} \to X$ factors through f.*

⋆ **Finiteness**

Proposition 6.12. *Let $f : X \to Y$ be a proper and affine morphism. Then f is finite. In fact, this is a corollary of the fact that a universally closed and affine morphism is integral.*

Proposition 6.13. *Let $f : X \to Y$ be a proper and quasi-finite morphism. Then f is finite*

Proposition 6.14. *Let $f : X \to Y$ be a finite morphism of varieties, $\mathcal{F}$ coherent sheaf on X. Then we have*
(1)$f_* \mathcal{F}$ is coherent and $R^i f_* \mathcal{F} = 0$ for all $i > 0$.
(2)$H^i(X, \mathcal{F}) = H^i(Y, f_* \mathcal{F})$ for all i. In particular, $\chi(X, \mathcal{F}) = \chi(Y, f_* \mathcal{F})$.

⋆ **Projectivity**

Since there are many different definitions of projective morphisms, to be clarified, we would call the quasi-coherent sheaf version definition weakly projective morphism, the vector bundle version definition strongly projective morphism and the Hartshorne's definition projective morphism. And prefix "quasi "would be used to mean the definitions where closed immersion is replaced by immersion.

Proposition 6.15. *Let $f : X \to Y$ be a proper morphism. If there is f-relative very ample line bundle on X, then f is weakly projective.*

Remark 6.3. *If moreover Y satisfies some resolution property that for any quasi-coherent sheaf $\mathcal{F}$ on Y, there is a surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$ for some locally free sheaf $\mathcal{E}$, then f is strongly projective. This remark also makes sense for many results in the following.*

Proposition 6.16. *Let $f : X \to Y$ be a morphism. Assume Y admits some ample line bundle. Then f is projective if and only if f is strongly projective if and only if f is weakly projective.*

Proposition 6.17. *Let $f : X \to Y$ be a finite morphism. Then f is weakly projective.*

Remark 6.4. *In general, finite morphism is neither projective nor strongly projective.*

Proposition 6.18. *Let $f : X \to Y$ be a morphism of finite type, Y quasi-compact. Assume $\mathcal{L}$ is an f-relative ample line bundle on X. Then $\mathcal{L}^{\otimes d}$ is f-relative very ample for d large enough.*

⋆ Flatness

Proposition 6.19. *Let $f : X \to Y$ be a morphism of schemes. Then f is flat if and only if for each $x \in X$, the local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$.*

Proposition 6.20. *Let $f : X \to Y$ be a flat morphism, Y irreducible with generic point η. Assume the fiber of f at η is irreducible. Then X is also irreducible.*

Proposition 6.21. *Let $f : X \to Y$ be a flat morphism locally of finite presentation. Then f is a universally open map.*

Corollary 6.5. *Let X be a scheme, $\mathcal{E}$ vector bundle on X of rank $n+1$. Denote $f : \mathbb{P}(\mathcal{E}) \to X$ the natural projection. Then f is a universally open map.*

⋆ Faithful Flatness

Proposition 6.22 (***Faithfully Flat Descent***)**.** *Let $f : X \to Y$ be a faithfully flat morphism of schemes. Then for all map $\varphi : \mathcal{F} \to \mathcal{G}$ between quasi-coherent sheaves on Y, φ is an isomorphism if and only if $f^*\varphi$ is an isomorphism.*

Corollary 6.6. *Let $f : X \to Y$ be a faithfully flat morphism of schemes. Then for each morphism $\varphi : T \to S$ between schemes affine over Y, φ is an isomorphism if and only if $\varphi \times_Y X$ is an isomorphism.*

Remark 6.5. *In fact, if moreover assume f is quasi-compact, there are lots of properties of morphisms that can be checked after base change by f, which is called descent property local in the fpqc topology. One can refer to* Stacks Project 02YJ *for more details.*

6.5 Dimension theory

Proposition 6.23. *Let X be an irreducible affine scheme. Then for any nonempty open subset $U \subseteq X$, we have that $\dim U = \dim X$.*

Remark 6.6. *This is not true for general topological space.*

Proposition 6.24. *Let X be a noetherian topological space, $Y \subseteq X$ constructible set. Then* $\dim Y = \dim \overline{Y}$.

Proposition 6.25. *Let X be an irreducible scheme locally of finite type over field k. Then for any field extension K/k, we have $X \times_k K$ is of pure dimension of* $\dim X$.

Corollary 6.7. *Let X be a scheme locally of finite type over field k. Then for any field extension K/k, we have* $\dim(X \times_k K) = \dim X$.

Corollary 6.8. *Let X be a k-scheme, K/k finite extension. Then* $\dim(X \times_k K) = \dim X$.

6.6 Universal injectivity

Proposition 6.26. *Let $f : \operatorname{Spec} L \to \operatorname{Spec} K$ be a morphism of schemes. Then f is universally injective if and only if field extension L/K is purely inseparable.*

Proposition 6.27. *Let $f : X \to Y$ be a morphism of schemes. Then f is universally injective if and only if $\Delta_{X/Y} : X \to X \times_Y X$ is surjective.*

6.7 Vector bundles

Proposition 6.28. *Let $\mathbb{P}^n_k$ be a projective space. Then there is a canonical inclusion* $\mathcal{O}_{\mathbb{P}^n_k}(-1) \hookrightarrow \mathcal{O}_{\mathbb{P}^n_k}^{\oplus n+1}$.

Corollary 6.9. *Let X be a scheme, $\mathcal{E}$ a vector bundle on X with associated projective bundle $\mathbb{P}(\mathcal{E})$. Then there is a canonical inclusion $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}$, where $\pi : \mathbb{P}(\mathcal{E}) \to X$ is the natural morphism.*

Proposition 6.29. *Let X be a scheme, $\mathcal{E}$ a vector bundle on X with associated projective bundle $\mathbb{P}(\mathcal{E})$. Then $\mathbb{P}(\mathcal{E})$ is irreducible if and only if X is irreducible.*

6.8 Exceptional locus

Proposition 6.30. *Let $f : X \to Y$ be a birational morphism between reduced and separated schemes, $g : Y \dashrightarrow X$ the birational inverse of f. Then $\{x \in X \big| g \text{ is not defined at } f(x)\}$ and $X \setminus U_X$ are the same set, denoted by* $\operatorname{Ex}(f)$*, where U_X denotes the maximal open dense subset of X such that $f\big|_{U_X}$ is an isomorphism.*

Proposition 6.31. *Let $f : X \dashrightarrow Y$ be a birational map between reduced, noetherian and separated schemes, $g : Y \dashrightarrow X$ the birational inverse of f. Then for any prime divisor D on Y not contained in* $\operatorname{Sing} Y$*, the generic point η of D is contained in U_Y.*

6.9 Ramification theory

Proposition 6.32. *Let $f : X \to Y$ be a finite morphism between smooth algebraic curves of degree d. Assume $P_1, \cdots, P_r$ are all preimages of $Q \in Y$. Then*

$$\sum_{i=1}^{r} e_{P_i} = d \tag{6}$$

6.10 Intersection theory

Proposition 6.33. *Let X be a scheme over field k, D Cartier divisor on X. If K/k is a field extension, $X_K := X \times_k K$ with projection $p : X_K \to X$, then for all $Z \in Z_d(X)$, $\deg_{p^*D} p^*Z = \deg_D Z$.*

7 Homological Algebra

Proposition 7.1. *Let $\mathscr{A}$ be an abelian category. Assume $I^\bullet$ be an acyclic bounded-below complex of injective objects. Then $I^\bullet$ is contractible i.e. $\mathrm{id}_{I^\bullet}$ is null-homotopic. Dually, acyclic bounded-above complex of projective objects is contractible.*

8 Proofs

8.1 Abstract Algebra 抽象代数

Proposition 1.1. $b = b1 = bac = 1c = c$ □

Proposition 1.2. If there exists $z \notin R$, while $z \in \cap_\alpha R_{\mathfrak{m}_\alpha}$, consider the ideal $I = \{x \in R \big| xz \in R\}$. Since $1 \notin I$, we have I is a proper ideal of R, thus there is a maximal ideal $\mathfrak{m}$ s.t. $I \subset \mathfrak{m}$. However, it is obvious that $z \notin R_\mathfrak{m}$, contradiction! □

Proposition 1.1. See in the website, Schur's Theorem on commutator subgroup. □

Proposition 1.3. Take normal cyclic subgroup H s.t. the index of H is minimal. Consider the centralizer C of H, and use Theorem 1.1 to show $C = H$. Finally, suppose there is an element $g \notin H$, try to make contradiction. □

Proposition 1.4. Consider the double coset for H. □

Proposition 1.5. Abelian case by Fundamental Theorem of Finitely Generated Abelian Groups and nonabelian case by Sylow Theorem. □

Proposition 1.6. By induction. □

Proposition 1.7. It suffices to prove the case for p-group. G is a p-group of order p^n, denote the number of maximal subgroups of G by m_G. By induction, we assume that $m_G \equiv 1 \pmod p$ for $\forall 1 \le n \le k$.

Now consider the $k+1$ case. If $m_G = 1$, then done. Otherwise, assume P, P' are two different maximal subgroups of G. Note that by Proposition 1.6, we have $P \trianglelefteq G$, $P' \trianglelefteq G$ and $PP' = G$. In addition, $PP'/P \cong P'/P \cap P'$. Thus $\sharp(P \cap P') = p^{k-1}$. For each maximal subgroup H of P, P' such that $P' \cap P = H$ one-to-one corresponds to a subgroup of G/H of order p except P/H. The correspondence gives a bijection. Thus $m_G \equiv 1 + \sum_{\text{H maximal subgroup of P}} (m_{G/H} - 1) \equiv 1 \pmod p$. □

Proposition 1.8. Obviously, G has no proper subgroup of index less than 5. Similarly, by Sylow's Theorem, the number of Sylow 3-groups and Sylow 5-groups are 10 and 6 respectively. And the number of Sylow 2-groups is 5 or 15. It is obvious to show $G \cong A_5$ in the first case. For the other case, there must exists two different Sylow 2-groups P and Q such that $P \cap Q \neq \{1\}$. Assume that $\langle a \rangle = P \cap Q$, then $o(a) = 2$. Consider the order of $C_G(a)$. □

Proposition 1.9. For any $g \in G$, $g^{-1}Pg \subseteq g^{-1}Ng = N$. Thus $g^{-1}Pg$ is still some Sylow p-group of N. By Sylow's Theorem, there exists $n \in N$ such that $g^{-1}Pg = n^{-1}Pn$. Then $g = (gn^{-1})n$, where $gn^{-1} \in N_G(P)$. □

8.2 Topology 拓扑

8.2.1 Basic topology

Proposition 2.1. "⇒": obviously. "⇐": consider $U_\alpha \setminus S$ □

8.2.2 Irreducibility

Proposition 2.2. Suppose that U is not dense, then there exist $x \in X$ and open neighbourhood $W \ni x$ such that $W \cap U = \varnothing$, contradiciting to irreducibility. Hence U is dense in X. □

Proposition 2.3. Suppose V is a maximal irreducible subspace of X. Consider the closure of V. Suffice to show $\overline{V}$ is closed. □

Proposition 2.4. Assume there is infinitely many irreducible components. Consider the descending chain of closed subsets

$$V_1 \supseteq V_1 \cap V_2 \supseteq \cdots \supseteq V_1 \cap \cdots \cap V_n \supseteq \cdots \tag{7}$$

where V_i are distinct irreducible components. □

8.2.3 dimension theory

8.3 Commutative Algebra 交换代数

8.3.1 Prime ideals

⋆ Radical

Proposition 3.1. See in the Atiyah. □

Proposition 3.2. See in the Atiyah. □

Proposition 3.3. $r(I \cup J) = r(r(I) \cup r(J)) = r(1) = (1)$ □

Proposition 3.4. Assume $\overline{x}$ is the idempotent in $A/\mathfrak{N}$, we have $x^2 - x$ is nilpotent. Thus $\exists n \in \mathbb{N}_+$ s.t. $x^n(1-x)^n = 0$. As (x) and (1 - x) are coprime, we have r(x) and r(1 - x) are coprime. Note that $r(x^n) = r(x)$ and $r((1-x)^n) = r(1-x)$, by Proposition 3.3, get (x^n) and $((1-x)^n)$ are coprime. By Chinese Remainder Theorem, we have

$$A = A/(0) = A/(x^n(1-x)^n) \cong A/(x^n) \times A/((1-x)^n) \tag{8}$$

Take e be the preimage of $(0,1) \in A/(x^n) \times A/((1-x)^n)$, e is the lifting of x. □

⋆ Associated Prime Ideals

Proposition 3.5. Consider the set $\Sigma := \{\mathrm{ann}(a) \text{ for some } a \big| \mathrm{ann}(a) \subseteq \mathfrak{p}\}$. Note that $0 = \mathrm{ann}(1)$ is an element in Σ. Then by noetherian condition, we get there is maximal element $\mathrm{ann}(a_0)$ in Σ. Want to show $\mathrm{ann}(a_0)$ is a prime ideal.

Suppose $bc \in \mathrm{ann}(a_0)$ and neither b nor c in $\mathrm{ann}(a_0)$. Since $bc \in \mathfrak{p}$, we may assume $b \in \mathfrak{p}$. Consider $\mathrm{ann}(a_0c)$ strictly containing $\mathrm{ann}(a_0)$. Then $\mathrm{ann}(a_0c) \not\subset \mathfrak{p}$ so that there is $b' \in \mathrm{ann} \setminus \mathfrak{p}$. Again consider $\mathrm{ann}(a_0b')$ strictly containing $\mathrm{ann}(a_0)$. And for any $c' \in \mathrm{ann}(a_0b')$, $b'c' \in \mathrm{ann}(a_0)$ so that $c' \in \mathfrak{p}$. Thus $\mathrm{ann}(a_0) \subsetneq \mathrm{ann}(a_0c') \subseteq \mathfrak{p}$, contradiciting to our choice of a_0. Hence $\mathrm{ann}(a_0)$ is a prime ideal contained in $\mathfrak{p}$. While $\mathfrak{p}$ is minimal, we get $\mathrm{ann}(a_0) = \mathfrak{p}$. In particular, if we take $\Sigma' := \{\mathrm{ann}(a) \text{ for some } a \notin \mathfrak{p} \big| \mathrm{ann}(a) \subseteq \mathfrak{p}\}$, same argument shows that we can take $a_0 \notin \mathfrak{p}$. □

Proposition 3.6. Consider the homomorphism $\varphi : R \to \prod_{i=1}^n R_{\mathfrak{p}_i}$ sending x to $(\frac{x}{1})_{1 \le i \le n}$. For injectivity, assume $\varphi(x) = 0$. Then $\mathrm{ann}(x) \not\subset \mathfrak{p}_i$ for all i so that $\mathrm{ann}(x) = R$ and hence $x = 0$.

For surjectivity, suffices to show $(0, \cdots, 0, \frac{x}{s}, 0, \cdots, 0) \in \mathrm{im}\,\varphi$ for all $x \in R$ and $s \notin \mathfrak{p}_i$. By Proposition 3.5, for each $\mathfrak{p}_i$, there is some element $a \notin \mathfrak{p}_i$ such that $\mathrm{ann}(a) = \mathfrak{p}_i$. Then for all $s \notin \mathfrak{p}_i$, we know $as \notin \mathfrak{N}$ so that by Proposition 3.2, there exists $b \in R$ such that $1 - abs$ is not invertible so that $\frac{1}{s} = \frac{ab}{1}$ in $R_{\mathfrak{p}_i}$. Hence $abx \mapsto (0, \cdots, 0, \frac{x}{s}, 0, \cdots, 0)$. Conclude that φ is isomorphic. □

8.3.2 Localization

Proposition 3.7. Consider natural map $f : R \longrightarrow \cap_{\mathfrak{m}\in Max(R)} R_{\mathfrak{m}}$. Want to show that f is surjective. It suffices to prove that $f_{\mathfrak{m}} : R_{\mathfrak{m}} \longrightarrow (\cap_{\mathfrak{m}\in Max R} R_{\mathfrak{m}})_{\mathfrak{m}}$ is surjective for each maximal ideal $\mathfrak{m}$. □

Proposition 3.8. Assume $\frac{a}{1} \in S^{-1}R$ and $\frac{a^n}{1} = 0$ for some n. Then there exists $s \in S$ such that $a^n s = 0$ and hence $(as)^n = 0$. While R is reduced, we know $as = 0$ so that $\frac{a}{1} = 0$ in $S^{-1}R$. Conclude that $S^{-1}R$ is reduced. □

Corollary 3.1. By Proposition 3.8, we know $R_{\mathfrak{p}}$ is reduced. While $\mathfrak{p}$ is minimal, the unique prime ideal of $R_{\mathfrak{p}}$ is then 0. Conclude that $R_{\mathfrak{p}}$ is a field. □

8.3.3 Modules

⋆ Finitely Generatedness

Proposition 3.9. See in the Atiyah. □

Proposition 3.10. For every nonzero $m \in M$, consider the annihilator $\operatorname{ann}_R(m)$. Note that the kernel of homomorphism $R \to Rm$ sending 1 to m is $\operatorname{ann}_R(m)$. Then $Rm \cong R/\operatorname{ann}_R(m)$. By noetherian condition, we can take maximal element $\operatorname{ann}_R(m_0)$ in $\Sigma := \{\operatorname{ann}_R(m) \big| m \neq 0 \in M\}$.

Want to show $\operatorname{ann}_R(m_0)$ is a prime ideal of R. Suppose there is $ab \in \operatorname{ann}_R(m_0)$ such that $a, b \notin \operatorname{ann}_R(m_0)$. Then $am_0 \neq 0$ and $bm_0 \neq 0$. Note that $\operatorname{ann}_R(am_0) \supseteq \operatorname{ann}_R(m_0)$, by maximality of $\operatorname{ann}_R(m_0)$, we get $\operatorname{ann}_R(am_0) = \operatorname{ann}_R(m_0)$ so that $b \in \operatorname{ann}_R(m_0)$, contradiction! Hence $\operatorname{ann}_R(m_0)$ is a prime ideal.

Set $M_1 = Rm_0$ and consider $M' = M/M_1$. Repeating this process, we get a filtration $0 = M_0 \supset M_1 \subset M_2 \subset \cdots$. While a finite module over noetherian ring is noetherian, hence the filtration is stationary. Assume it stops at M_n. Remains to show $M_n = M$. In fact, by construction, we must have that $M/M_n = 0$ otherwise we would have $M_n \subsetneq M_{n+1}$. □

Corollary 3.2. By Proposition 3.10, there is a filtration

$$0 = I_0 \subset I_1 \subset \cdots \subset I_n = R \tag{9}$$

such that $I_i/I_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$. Assume $\mathfrak{q}$ appears m times. Note that $0 \to I_{i-1} \to I_i \to R/\mathfrak{p}_i \to 0$ is exact, tensoring by $R_{\mathfrak{q}}$, we get exact sequences

$$0 \longrightarrow I_{i-1} \otimes_R R_{\mathfrak{q}} \longrightarrow I_i \otimes_R R_{\mathfrak{q}} \longrightarrow R/\mathfrak{p}_i \otimes_R R_{\mathfrak{q}} \longrightarrow 0 \tag{10}$$

While for any $\mathfrak{p}_i \neq \mathfrak{q}$, $R/\mathfrak{q}_i \otimes_R R_{\mathfrak{q}} = 0$ and $R/\mathfrak{q} \otimes_R R_{\mathfrak{q}} = \operatorname{Frac}(R/\mathfrak{q})$. Summing on length over $R_{\mathfrak{q}}$, we get $m = \operatorname{length} R_{\mathfrak{q}}$. □

⋆ Flatness

Proposition 3.11. See in the website Stacks Project 00NX. □

8.3.4 Algebras

⋆ Finite Generatedness

⋆ Integrality

Proposition 3.12. Since L/K is algebraic, for all $\alpha \in L$, there is a minimal polynomial $x^n + \frac{a_1}{a_1'}x^{n-1} + \cdots + \frac{a_n}{a_n'}$, where all a_i and a_i' are in A. Take the multiplication a of all dominators, then $a\alpha$ is a root of $x^n + \frac{a_1 a}{a_1'}x^{n-1} + \cdots \frac{a_n a^n}{a_n'}$ with all coefficients in A. By definition, we know $a\alpha \in B$ and hence $\alpha \in \operatorname{Frac} B$. In conclusion, $\operatorname{Frac} B = L$. □

Proposition 3.13. By Proposition 3.12, we only need to show that for all $b \in B$, $b^{-1} \in B \otimes_A K$. In fact, $B \otimes_A K = S^{-1}B$ where $S = A \setminus \{0\}$. And for all $b \in B$, there is a minimal polynomial $x^n + a_1x^{n-1} + \cdots + a_n$ with coefficients in A. By definition, we know $a_n \neq 0$. Denote $c = b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1}$, we get that $bc = -a_n \in A$ so that $b^{-1} = \frac{c}{-a_n}$. □

Proposition 3.14. When B is a field, since A is a subring of B, A is also a field. Conversely, assume A is a field. As B is integral over A, for all nonzero $b \in B$, there is a minimal monic polynomial $x^n + a_1x^{n-1} + \cdots + a_n$ of b over A. Since B is integral domain, we know $a_n \neq 0$, otherwise the polynomial is not minimal. Then $b(b^{n-1} + a_1b_{n-2} + \cdots + a_{n-1}) \in A^\times$ so that b is invertible. Conclude that B is a field. □

Corollary 3.3. Since localization is exact and integrality is stable under localization, we may assume $\mathfrak{p}$ is maximal by relacing A and B by $A_\mathfrak{p}$ and $B_\mathfrak{p}$ respectively. Suppose $\mathfrak{P}_1 \subset \mathfrak{P}_2$. Take $A' = A/\mathfrak{p}$ and $B' = B/\mathfrak{P}_1$, then there is an integral extension $A' \hookrightarrow B'$ of integral domains. Note that A' is a field, by Proposition 3.14, we get B' is a field so that $\mathfrak{P}_2/\mathfrak{P}_1 = 0$ and hence $\mathfrak{P}_1 = \mathfrak{P}_2$, contradiction! □

Corollary 3.4. Immediately comes from Corollary 3.3. □

Corollary 3.5. Immediately comes from Corollary 3.4. □

Proposition 3.15. Denote $f : A \to B$. Assume $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ are all minimal prime ideals of A. Then

$$\widetilde{A} = \prod_{i=1}^{n} \overline{A/\mathfrak{p}_i} \tag{11}$$

where $\overline{A/\mathfrak{p}_i}$ denotes its integral closure in fraction field $k(\mathfrak{p}_i) = \operatorname{Frac}(A/\mathfrak{p}_i)$. By hypothesis, for each minimal prime $\mathfrak{p}_i$, there is some minimal prime $\mathfrak{q}_i$ of B lying over $\mathfrak{p}_i$ with the fraction field extension isomorphic.

Since $B/\mathfrak{q}_i$ is integral over $A/\mathfrak{p}$, its image in $k(\mathfrak{p}_i)$ is contained in the integral closure $\overline{A/\mathfrak{p}}$. Hence we have a natural inclusion

$$B/\mathfrak{q}_i \hookrightarrow \overline{A/\mathfrak{p}}. \tag{12}$$

Define a ring homomorphism

$$\varphi : B \longrightarrow \prod_{i=1}^{n} B/\mathfrak{p}_i \longrightarrow \widetilde{A} = \prod_{i=1}^{n} \overline{A/\mathfrak{p}_i} \tag{13}$$

sending b to $(\bar{b})_{1\leq i\leq n}$. Remains to check that the composition map $\varphi\circ f$ is just the orginal normalization map. Consider the commutative diagrams

$$\begin{tikzcd} A \arrow[r] \arrow[d, "f"] & A/\mathfrak{p}_i \arrow[r, hook] \arrow[d, "\bar{f}"] & k(\mathfrak{p}_i) \arrow[d, "\sim"] \\ B \arrow[r] & B/\mathfrak{q}_i \arrow[r, hook] & k(\mathfrak{q}_i) \end{tikzcd} \tag{14}$$

Hence for all $a\in A$, $\overline{f(a)}=\bar{a}$ in $\overline{A/\mathfrak{p}_i}$, done! □

⋆ Finiteness

Proposition 3.16. i don't know □

⋆ Flatness

Proposition 3.17. "$\Rightarrow$ ": Since B is flat over A, then $B\otimes_A A_\mathfrak{p}$ is flat over $A_\mathfrak{p}$. Note that $A_\mathfrak{p}\to B_\mathfrak{P}$ factors through $B\otimes_A A_\mathfrak{p}$ and $B_\mathfrak{P}$ is localization of $B\otimes_A A_\mathfrak{p}$. Hence $B_\mathfrak{P}$ is flat over $A_\mathfrak{p}$.

"$\Leftarrow$ ": Given a short exact sequence $0\to M'\to M\to M''\to 0$. Suffices to show $0\to M'\otimes_A B_\mathfrak{P}\to M\otimes_A B_\mathfrak{P}\to M''\otimes_A B_\mathfrak{P}\to 0$ exact for all $\mathfrak{P}\in\operatorname{Spec}B$. While $M\otimes_A B_\mathfrak{P}\cong M\otimes_A A_\mathfrak{p}\otimes_{A_\mathfrak{p}} B_\mathfrak{P}$, done! □

Proposition 3.18. i don't know. □

Proposition 3.19. See in the website, Stacks Project 00HR. □

⋆ Faithful Flatness

Proposition 3.20. Suffice to show that for any A-module M, if $M\otimes_A B=0$, then $M=0$. Assume that there is a nonzero element $m\in M$, consider $\operatorname{ann}(m)$. Then $Am\cong A/\operatorname{ann}(m)$ is a nonzero submodule of M. As B is flat over A, we have $A/\operatorname{ann}(m)\otimes_A B\cong B/\operatorname{ann}(m)B$ is a submodule of $M\otimes_A B$. While there exists prime ideal $\mathfrak{p}\in\operatorname{Spec}A$ containing $\operatorname{ann}(m)$ and there is a prime ideal $\mathfrak{P}\in\operatorname{Spec}B$ lying over $\mathfrak{p}$, we have $\operatorname{ann}(m)B\subseteq\mathfrak{P}\neq B$. Thus $B/\operatorname{ann}(m)B\neq 0$, contradiction! □

Corollary 3.6. Note that $IB=I\otimes_A B$. This immediately comes from Proposition 3.20. □

Corollary 3.7. Consider kernel and cokernel of φ. There is an exact sequence

$$0\longrightarrow\ker\varphi\longrightarrow\mathcal{F}\xrightarrow{\varphi}\mathcal{G}\longrightarrow\operatorname{coker}\varphi\longrightarrow 0 \tag{15}$$

Tensoring by B, we get an exact sequence

$$0\longrightarrow\ker\varphi\otimes_A B\longrightarrow\mathcal{F}\otimes_A B\xrightarrow{\varphi\otimes_A B}\mathcal{G}\otimes_A B\longrightarrow\operatorname{coker}\varphi\otimes_A B\longrightarrow 0 \tag{16}$$

where by assumption $\varphi\otimes_A B$ is an isomorphism. By Proposition 3.20, we get $\ker\varphi=0$ and $\operatorname{coker}\varphi=0$, done! □

8.3.5 Dimension theory

Proposition 3.21. For any $x \neq 0 \in R$, consider the descending chain of ideals

$$(x) \supseteq (x^2) \supseteq \cdots \supseteq (x^n) \supseteq \cdots \tag{17}$$

By Artinian property, we can get that x is invertible. □

Proposition 3.22. As reduction does not change topology, we may assume A is integral domain, denote $F := \mathrm{Frac}(A)$. Then by Theorem 1.8A in Hartshorne Chapter I, we know $\dim A = \mathrm{trdeg}(F/k)$. Assume $\dim A = d$ and $x_1, \cdots, x_d \in F$ transcandence basis of F over k. Denote $E := k(x_1, \cdots, x_d)$, then F is algebraic over E.

Note that $A \otimes_k K$ is flat over A, by Going-down Theorem for flat homomorphism, for every minimal prime ideal $\mathfrak{p}$ of $A \otimes_k K$, it would corresponds to a minimal prime ideal $\mathfrak{P}$ of $F \otimes_k K$ and again by Theorem 1.8A, we have

$$\dim \overline{\{\mathfrak{p}\}} = \mathrm{trdeg}(\mathrm{Frac}(A \otimes_k K/\mathfrak{p})/K) = \mathrm{trdeg}(\mathrm{Frac}(F \otimes_k K/\mathfrak{P})/K) \tag{18}$$

Denote $L := \mathrm{Frac}(F \otimes_k K/\mathfrak{P})$. Note that $F \otimes_k K$ is integral over $E \otimes_k K$ and the latter one is isomorphic to a localization of $K[x_1, \cdots, x_d]$ which is an integral domain. Also as $F \otimes_k K$ is flat over $E \otimes_k K$, we get $\mathfrak{P}$ contracts to (0) in $E \otimes_k K$. Hence there is an injection

$$E \otimes_k K \hookrightarrow F \otimes_k K \twoheadrightarrow F \otimes_k K/\mathfrak{P} \hookrightarrow L \tag{19}$$

Hence $\mathrm{trdeg}(L/K) = \mathrm{trdeg}(\mathrm{Frac}(E \otimes_k K)/K) = \mathrm{trdeg}(K(x_1, \cdots, x_d)/K) = d$, done! □

Corollary 3.8. Apply Proposition 3.22 to $A_{\mathfrak{p}}$ for every minimal prime ideal $\mathfrak{p}$ of A. □

Corollary 3.9. First note that base change preserve surjective morphism, we get $\mathrm{Spec}(A \otimes_k K) \to \mathrm{Spec}\, A$ is surjective. Then for any maximal ideal of A, its preimage in $\mathrm{Spec}(A \otimes_k K)$ is nonempty. Hence for any strictly increasing sequence of prime ideals of A, by Going-down Theorem for flat homomorphism, it can be lifted to a strictly increasing sequence of prime ideals of $A \otimes_k K$ so that $\dim A \leq \dim(A \otimes_k K)$.

Conversely, for any $\mathfrak{P}_1 \subsetneq \mathfrak{P}_2$ in $A \otimes_k K$, want to show their contractions in A are distinct. Suppose $\mathfrak{P}_1$ and $\mathfrak{P}_2$ both lie over $\mathfrak{p} \in \mathrm{Spec}\, A$. Since K is flat over k, $A/\mathfrak{p} \otimes_k K \cong (A \otimes_k K)/(\mathfrak{p} \otimes_k K)$ so that $\mathfrak{P}_1$ and $\mathfrak{P}_2$ would corresponding to prime ideals in $A/\mathfrak{p} \otimes_k K$. Hence we amy assume A is integral domain and $\mathfrak{p} = (0)$.

Consider $\mathrm{Frac}(A) = A_{\mathfrak{p}}$ and $\mathrm{Frac}(A) \otimes_k K$. We get $\mathfrak{P}_1$ and $\mathfrak{P}_2$ would correspondend prime ideals in $\mathrm{Frac}(A) \otimes_k K$. Now by Proposition 3.22, as K/k is finite, we have $\dim(\mathrm{Frac}(A) \otimes_k K) = \dim K = 0$, contradiciting to $\mathfrak{P}_1 \subsetneq \mathfrak{P}_2$. Thus strictly increasing sequence of prime ideals of $A \otimes_k K$ would restrict to strictly increasing sequence of prime ideals of A. Conclude that $\dim(A \otimes_k K) = \dim A$. □

8.3.6 Length

8.3.7 Local Rings

Proposition 3.23. Assume that $\mathfrak{m}/\mathfrak{m}^2$ is spanned by $\overline{a_1}, \cdots, \overline{a_n}$ for $a_i \in R$. Get $\mathfrak{m} = (a_1, \cdots, a_n) + \mathfrak{m}^2$. By Nakayama's Lemma, we have $\mathfrak{m} = (a_1, \cdots, a_n)$. □

Corollary 3.10. Immediately comes from Proposition 3.23. $\square$

Proposition 3.24. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) is clear. Suffices to prove (4) $\Rightarrow$ (1). As M is finitely generated, by Nakayama's Lemma, we can pick representative elements of a basis of $M/\mathfrak{m}M$ over k, which also generate M. And these elements induces a surjection $R^{\oplus n} \twoheadrightarrow M$ with kernel N. Since $\mathrm{Tor}_1^R(M,k)=0$, the following sequence is exact

$$0 \longrightarrow N \otimes_R k \longrightarrow R^{\oplus n} \otimes_R k \longrightarrow M \otimes_R k \longrightarrow 0 \tag{20}$$

Note that by our choice, $R^{\oplus n} \otimes_R k \to M \otimes_R k$ is an isomorphism. Hence $N \otimes_R k = 0$ so that by Nakayama's Lemma again, $N = 0$. Conclude that $M \cong R^{\oplus n}$ is free. $\square$

Corollary 3.11. Immediately comes from Proposition 3.24 and the fact that for a finite module over a PID is free if and only if it is torsion-free. $\square$

⋆ Valuation Rings

Proposition 3.25. We may assume $v(a) < v(b)$ so that $v(a+b) \geq v(a)$. Again by the ultrametric inequality, $v(a) \geq \min\{v(a+b), v(-b)\}$. Since $v(-b) = v(b) > v(a)$, we get $v(a) \geq v(a+b)$, done! $\square$

Proposition 3.26. Replacing A by $A/\mathfrak{p}$, we may assume A is an integral domain and $\mathfrak{p}=0$. Also by replacing A by $A_{\mathfrak{P}}$, we may assume A is a local ring and $\mathfrak{P}$ is the maximal ideal of A. Let $K = \mathrm{Frac}(A)$ be the fraction field of A. When $\dim A = 0$ i.e. $A = K$ is a field, then we can embed K into $K[[t]]$.

When $\dim A \geq 1$, consider the graded ring $G = \bigoplus_{n\geq 0} \mathfrak{P}^n/\mathfrak{P}^{n+1}$. Since A is noetherian, G is a finitely generated graded algebra over the residue field $A/\mathfrak{P}$, and it is generated by the degree-1 piece $G_1 := \mathfrak{P}/\mathfrak{P}^2$. If every element of G_1 were nilpotent, then the entire irrelevant ideal G_+ would be nilpotent. This would imply $\mathfrak{P}^N = \mathfrak{P}^{N+1}$ for some integer N. By Nakayama's Lemma, this forces $\mathfrak{P}^N = (0)$, which is impossible since $\dim A \geq 1$. Therefore, there exists an element $x \in \mathfrak{P}$ such that $\overline{x}$ is not nilpotent.

Construct $B := A[\frac{\mathfrak{P}}{x}] \subset K$. Then B is also a noetherian integral domain. It is clear that $xB = \mathfrak{P}B$. Want to show $xB \neq B$ is a proper principal ideal of B. Suppose that $xB = B$. Since $xB = \mathfrak{m}B$, we would have $1 \in \mathfrak{P}B$. Thus we could write

$$1 = \sum_i c_i b_i = \sum_{i,j} c_i \cdot \frac{a_{ij}}{x^{k_{ij}}} \tag{21}$$

where $c_i \in \mathcal{P}$ and $b_i = \sum_j \frac{a_{ij}}{x^{k_{ij}}} \in B$. Multiplying by x^N for large enough N yields

$$x^N = \sum_{i,j} c_i a_{ij} x^{N-k_{ij}} \in \mathfrak{m} \cdot \mathfrak{m}^N = \mathfrak{m}^{N+1}, \tag{22}$$

contradicting to our choice of x. Therefore xB is a proper ideal of B. Now by Krull's Principal Ideal Theorem, any minimal prime $\mathfrak{q}$ containing xB has height 1. Localizing B at $\mathfrak{q}$, we get a 1-dimensional local noetherian domain. Now, let C be the integral closure of $B_{\mathfrak{q}}$ in K. It is clear that C is again Noetherian and 1-dimensional so that C is a Dedekind domain.

Since C is a Dedekind domain integral over $B_{\mathfrak{q}}$, by the Going-up theorem we can pick a maximal ideal $\mathfrak{m}$ of C that contracts to the maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$. Take $R := C_{\mathfrak{m}}$, then C is a DVR and there is a sequence of ring homomorphism

$$A \hookrightarrow B \hookrightarrow B_{\mathfrak{q}} \hookrightarrow C \hookrightarrow R \tag{23}$$

Clearly, (0) in R contracts to (0) in A and under the spectrum map, we have

$$\mathfrak{m} \longmapsto \mathfrak{q}B_{\mathfrak{q}} \longmapsto \mathfrak{q} \longmapsto \mathfrak{P} \tag{24}$$

done! □

⋆ Artinian Local Rings

Proposition 3.27. Assume $\operatorname{length}(R) = l$. Take the sequence $0 = M_0 \subset M_1 \subset \cdots \subset M_l := R$ with exact sequences

$$0 \longrightarrow M_{i-1} \longrightarrow M_i \longrightarrow R/\mathfrak{m} \longrightarrow 0 \tag{25}$$

Counting dimension over k and summing over these short exact sequences, we get $\dim_k R = \dim_k M_l = \sum_i [R/\mathfrak{m} : k] = l \cdot [R/\mathfrak{m} : k]$. □

Proposition 3.28. By Corollary 3.8, $\dim(R \otimes_k K) = \dim R = 0$. Also $R \otimes_k K$ is of finite type over K and hence noetherian. Thus $R \otimes_k K$ is artinian. By Proposition 3.6, we get $R \otimes_k K \cong \prod_i R_i$, where R_i is the localization of R at minimal prime ideals hence artinian local ring.

Note that composition series of $R \otimes_k K$ can be constructed as following

$$0 \subset M_0^1 \subset M_1^1 \subset \cdots \subset M_{l_1}^1 = M_{l_1}^1 \oplus M_0^2 \subset M_{l_1}^1 \oplus M_1^2 \subset \cdots \tag{26}$$

where $l_i = \operatorname{length}(R_i)$ and $M_0^i \subset \cdots \subset M_{l_i}^i$ is composition series of R_i, with short exact sequences

$$0 \longrightarrow \oplus_{j=1}^{i-1} M_{l_j}^j \oplus M_k^i \longrightarrow \oplus_{j=1}^{i-1} M_{l_j}^j \oplus M_{k+1}^i \longrightarrow R_i/\mathfrak{m}_i \longrightarrow 0 \tag{27}$$

Counting dimension over K and summing over short exact sequences, we get

$$\dim_K(R \otimes_k K) = \sum_i \operatorname{length}(R_i) \cdot [R_i/\mathfrak{m}_i : K] \tag{28}$$

By Proposition 3.27, we know $\dim_k(R) = \operatorname{length}(R) \cdot [R/\mathfrak{m} : k]$, done! □

Proposition 3.29. By Corollary 3.9, $\dim(R \otimes_k K) = \dim R = 0$. Also $R \otimes_k K$ is finite over R and hence noetherian. Thus $R \otimes_k K$ is artinian. With same argument as Proposition 3.28, we are done! □

⋆ Henselian Local Rings

Proposition 3.30. Refer to Stacks Project 04GG. □

Corollary 3.12. Obviously □

Proposition 3.31. For any $\frac{a}{b} \in \mathrm{Frac}(R)\backslash R$ integral over R, assume that $\gcd(a,b) = 1$. There exists a monic polynomial $x^n + a_1x^{n-1} + \cdots + a_n$ where $a_i \in R$ s.t. $(\frac{a}{b})^n + a_1(\frac{a}{b})^{n-1} + \cdots + a_n = 0$. Thus $a^n = -b(a_1a^n + \cdots + a_nb^{n-1})$, get $b \mid a^n$ which contradicts to our assumption, $\gcd(a,b) = 1$. □

Proposition 3.32. Consider $f : A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{p}}$. Then the maximal ideal of $B_{\mathfrak{p}}$ lies over $\mathfrak{p}$. □

Proposition 3.33. By definition. □

Proposition 3.34. For any minimal prime ideal $\mathfrak{p}$ of A containing kernel, consider set $V = V(\varphi(\mathfrak{p}))$ in $\operatorname{Spec} B$, which by assumption is nonempty. As A, B are both noetherian, V is finite. Suppose that preimages of all $\mathfrak{P} \in V(\varphi(\mathfrak{p}))$ are not $\mathfrak{p}$, then each $\mathfrak{P}_i$ gives an $a_i \notin \mathfrak{p}$. Set $a = \prod_i a_i$, then $a \notin \mathfrak{p}$ and $\varphi(a) \in \cap_i \mathfrak{P}_i$ so that $\varphi(a) \in \sqrt{\varphi(\mathfrak{p})}$. Note that $\varphi^{-1}(\varphi(\mathfrak{p})) = \mathfrak{p}$, we get $a \in \mathfrak{p}$, contradiciton. □

Corollary 3.13. Consider ring homomorphism $\widetilde{\varphi} : A/\mathfrak{p} \to B/\varphi(\mathfrak{p})B$. Then apply Proposition 3.34, done! □

8.4 Linear Algebra 线性代数

Proposition 4.1. Consider the maximal linearly independent system □

8.5 Representation Theory 表示论

Theorem 5.1. See in the note. □

Theorem 5.2. Note that $\chi^{g_i} = \chi$, we have

$$\begin{aligned}
\langle Res_N^G(\chi), \varphi^{g_i}\rangle &= \langle Res_N^G(\chi^{g_i}), \varphi^{g_i}\rangle \\
&= \langle Res_N^G(\chi)^{g_i}, \varphi^{g_i}\rangle \\
&= \langle Res_N^G(\chi), \varphi\rangle \\
&= e
\end{aligned}$$

For the proof of showing φ^{g_i} are all the direct summand of $Res_N^G(\chi)$, see in the book Representation Theory of Finite Group Extensions. □

8.6 Algebraic Geometry

8.6.1 Classic examples

Proposition 6.1. See in the website, invertible sheaves of the projective line. □

8.6.2 Spectrum

Proposition 6.2. It is clear that for all $\mathfrak{p} \in \operatorname{Spec} B$, $\ker \varphi \subseteq \varphi^{-1}(\mathfrak{p})$ so that $f(\mathfrak{p}) \in V(\ker \varphi)$. Thus $\operatorname{im} f \subseteq V(\ker \varphi)$ □

Corollary 6.1. Immediately comes from Proposition 6.2 and Corollary 3.13. □

Corollary 6.2. Immediately comes from Proposition 6.2. □

Proposition 6.3. For different $\mathfrak{p}_1$ and $\mathfrak{p}_2$ in $\operatorname{Spec} B$, there must exists $a \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$ so that $\varphi^{-1}(a) \subseteq \varphi^{-1}(\mathfrak{p}_1) \setminus \varphi^{-1}(\mathfrak{p}_2)$ and hence $f(\mathfrak{p}_1) \neq f(\mathfrak{p}_2)$. □

Proposition 6.4. Clearly. □

8.6.3 Sheaves

⋆ Quasi-coherent Sheaves

Proposition 6.5. By Proposition 2.1, closedness is local. Hence we can reduce to affine case. Assume $X = \operatorname{Spec} A$ and $\mathcal{F} = \widetilde{M}$ for some finite A-module M. Pick generators $m_1, \cdots, m_n$ of M. Then $M_x = 0$ if and only if $(m_i)_x = 0$ for all i. Conclude that $\operatorname{Supp} \mathcal{F} = \cup_{i=1}^n \operatorname{Supp} m_i$ is closed. □

⋆ Flatness

Proposition 6.6. Suffices to check stalkwise. Denote the two projections to be $p : X_y \to X$ and $q : X_y \to \operatorname{Spec} k(y)$. For every point $x' \in X_y$, note that $(\mathcal{F}\big|_{X_y})_{x'} \cong \mathcal{F}_{p(x')} \otimes_{\mathcal{O}_{X,p(x')}} \mathcal{O}_{X_y,x'} \cong \mathcal{F}_{p(x')} \otimes_{\mathcal{O}_{Y,y}} k(y)$ for all sheaf module $\mathcal{F}$, we have exact sequence

$$\operatorname{Tor}_1^{\mathcal{O}_{Y,y}}(\mathcal{H}_{p(x')}, k(y)) \longrightarrow (\mathcal{F}\big|_{X_y})_{x'} \longrightarrow (\mathcal{G}\big|_{X_y})_{x'} \longrightarrow (\mathcal{H}\big|_{X_y})_{x'} \longrightarrow 0 \tag{29}$$

By assumption, $\mathcal{H}$ is flat over Y, hence $\operatorname{Tor}_1^{\mathcal{O}_{Y,y}}(\mathcal{H}_{p(x')}, k(y)) = 0$ and we are done! □

Proposition 6.7. Immediately comes from Proposition 3.11. □

Corollary 6.3. Immediately comes from Proposition 6.7. □

8.6.4 Properties of morphisms

⋆ Separatedness

Proposition 6.8. Since the problem is locally, we may assume $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are both affine. Consider the following commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "g"] & Y \arrow[d] \\ Y \arrow[r] & \operatorname{Spec} \mathbb{Z} \end{tikzcd} \tag{30}$$

By universal property of fibered product, we get a lifting $(f, g) : X \to Y \times Y$. Note that the set $\{x \in X \big| f(x) = g(x)\}$ is the preimage of Δ_Y under (f, g), which is a closed subset of X since Y is separated. As f and g agree on an open dense subset U of X, we get $U \subset \{x \in X \big| f(x) = g(x)\}$ and hence $\{x \in X \big| f(x) = g(x)\} = X$.

Now we have two ring homomorphisms $f^\sharp : B \to A$ and $g^\sharp : B \to A$ satisfying that for all $\mathfrak{p} \in \operatorname{Spec} A$, $f(\mathfrak{p}) = g(\mathfrak{p})$. Remains to show $f^\sharp = g^\sharp$. As X is reduced, it suffices to show that for all $b \in B$, $V(f^\sharp(b) - g^\sharp(b)) = X$. Since U is dense in X, it suffices to show that $U \subseteq V(f^\sharp(b) - g^\sharp(b))$.

In fact, since for all $\mathfrak{p} \in U$, we can take an affine open neighbourhood $D(s) \subseteq U$. Then $B \xrightarrow{f^\sharp} A \to A_s$ and $B \xrightarrow{g^\sharp} A_s$ are same. Hence we get $\frac{f^\sharp(b)}{1} = \frac{g^\sharp(b)}{1}$ in A_s and $(f^\sharp(b) - g^\sharp(b))s^n = 0$ for some n. As $s \notin \mathfrak{p}$, $f^\sharp(b) - g^\sharp(b) \in \mathfrak{p}$ and we are done! $\square$

Proposition 6.9. Consider the following commutative diagram

$$\begin{tikzcd} X_0 \arrow[r, "g'"] \arrow[dr, "\mathrm{id}"'] & Y_0 \times_X X_0 \arrow[r, hook] \arrow[d, "f'"] & Y_0 \arrow[r, hook, "j"] \arrow[d, "f \circ j"'] & Y \arrow[dl, "f"] \\ & X_0 \arrow[r, hook, "i"] & X & \end{tikzcd} \tag{31}$$

For convenience, we reduce the orginal setting to $X = X_0$ and $Y = Y_0 \times_X X_0$. To show f is isomorphic, suffices to prove affine locally, so assume $X = \operatorname{Spec} A$. Since f is separated, by Cancellation Theorem, we know g is a closed immersion so that $g^\sharp : \mathcal{O}_Y \to g_*\mathcal{O}_X$ is surjective. And by definition of schematic image, $g^\sharp$ is also injective and hence isomorphic. Thus the ideal sheaf of closed immersion g is (0). Conclude that g is isomorphic and so is f. $\square$

⋆ Properness

Proposition 6.10. See in the website Stacks Project 02O5. $\square$

⋆ Integrality

Proposition 6.11. Since the normalization is an affine morphism, the statement is local. First to construct the factorization affine locally and then glue the resulting morphisms. Affine locally, assume $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$. Then by Proposition 3.15, we see $\widetilde{X} \to X$ factors through f.

Since the construction is compatible with localization, these local morphisms glue together to give a global morphism of schemes

$$g : \widetilde{X} \longrightarrow Y \tag{32}$$

such that the following diagram commutes

$$\begin{tikzcd} \widetilde{X} \arrow[rr] \arrow[dr, "g"'] & & X \\ & Y \arrow[ur, "f"'] & \end{tikzcd} \tag{33}$$

Hence the normalization map factors through f. $\square$

Corollary 6.4. Consider $f' : Y^{\mathrm{red}} \to Y \to X$. Since reduction donot change topology and closed immersion is integral, f' is still dominant integral morphism. In addition, the generic stalk condition holds. Thus by Proposition 6.11, done! $\square$

⋆ **Finiteness**

Proposition 6.12. See in the website proper + affine = finite. □

Proposition 6.13. See in the website Stacks Project 02LS. □

Proposition 6.14. By Proposition 6.10, we see $f_*\mathcal{F}$ is coherent. For higher pushforward, since f is affine, we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$. Then by Serre Vanishing, $R^i f_*\mathcal{F} = H^i(\widetilde{X}, \mathcal{F}) = 0$ for all $i \geq 1$.

For the second result, since X and Y are varieties, we can compute cohomology with Čech complex. Take finite affine open covering $\mathcal{U} = \{U_i = \operatorname{Spec} A_i\}$ of Y. Assume $V_i := X \times_Y U_i$ to be $\operatorname{Spec} B_i$. Consider the Čech complexes

$$0 \longrightarrow \prod_i f_*\mathcal{F}(U_i) \longrightarrow \prod_{i,j} f_*\mathcal{F}(U_i \cap U_j) \longrightarrow \cdots \tag{34}$$

and

$$0 \longrightarrow \prod_i \mathcal{F}(V_i) \longrightarrow \prod_{i,j} \mathcal{F}(V_i \cap V_j) \longrightarrow \cdots \tag{35}$$

Note that $f_*\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_k}) = \mathcal{F}(V_{i_0} \cap \cdots \cap V_{i_k})$. It is clear that these two complexes are same so that we get same cohomology. □

⋆ **Projectivity**

Proposition 6.15. First f-relative very ample line bundle would give an immersion into $\mathbb{P}(\mathcal{F})$ for some quasi-coherent $\mathcal{F}$ on Y. While f is proper, hence immersion becomes closed immersion. □

Proposition 6.16. See in the website Stacks Project 087S. □

Proposition 6.17. For weakly See in the website Stacks Project 0B3I. □

Proposition 6.18. See in the website Stacks Project 01VU. □

⋆ **Flatness**

Proposition 6.19. "$\Leftarrow$ ": for all affine open subset $V = \operatorname{Spec} A$ of Y, and for all affine open subset $U = \operatorname{Spec} B$ of $f^{-1}(V)$, given a short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{36}$$

Tensoring with $A_\mathfrak{p}$ over A for all $\mathfrak{p} \in \operatorname{Spec} A$, we get a short exact sequence

$$0 \longrightarrow M'_\mathfrak{p} \longrightarrow M_\mathfrak{p} \longrightarrow M''_\mathfrak{p} \longrightarrow 0 \tag{37}$$

By assumption, $B_\mathfrak{P}$ is flat over $A_\mathfrak{p}$ for all $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$, we get a short exact sequence

$$0 \longrightarrow M'_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow M_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow M''_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow 0 \tag{38}$$

Note that $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} B_{\mathfrak{P}} = M \otimes_A B \otimes_B B_{\mathfrak{P}}$. Hence by local property of module homomorphism, we get a short exact sequence

$$0 \longrightarrow M' \otimes_A B \longrightarrow M \otimes_A B \longrightarrow M'' \otimes_A B \longrightarrow 0 \tag{39}$$

Conclude that B is flat over A so that f is flat.

"$\Rightarrow$ ": for any $x \in X$, consider the following commutative diagram

$$\begin{tikzcd} \operatorname{Spec} \mathcal{O}_{X,x} \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} \mathcal{O}_{Y,f(x)} \arrow[r] & Y \end{tikzcd} \tag{40}$$

By base change, we get a morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to \operatorname{Spec} \mathcal{O}_{Y,f(x)}$. By assumption, this morphism is flat and hence $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. $\square$

Proposition 6.20. For a generic point η' of X, assume $f(\eta') = y$. Take an affine open neighbourhood U of y and affine open neighbourhood $W \subseteq f^{-1}(U)$ of η'. Assume $U = \operatorname{Spec} A$ and $W = \operatorname{Spec} B$. Then B is flat over A. By Going-down Theorem, there exists some generalization x of η' mapping to η. While η' is generic point, this forces $y = \eta$. By assumption, the fiber at η is irreducible. This implies that X has unique generic point and hence irreducible. $\square$

Proposition 6.21. See in the website Stacks Project 01UA. $\square$

Corollary 6.5. Since f is smooth and proper, by Proposition 6.21, done! And there is a much more elementary way to prove this result.

Since openness is a local property, we can reduce to affine case. Especially, we can take affine open covering trivializing $\mathcal{E}$. Now f becomes $\mathbb{P}^n_A \to \operatorname{Spec} A$. Note that after base change, the morphism is still the projection from projective space. Suffices to show f is open map. In fact, we can moreover reduce to affine space $\mathbb{A}^n_A$.

Now for any closed subset $V = V(I) \subset \mathbb{A}^n_A$, we construct an ideal

$$I_A = \{a \in A \big| a \text{ appears in coefficients of some } f \in I\} \tag{41}$$

Note that for any prime ideal $\mathfrak{p} \in \operatorname{Spec} A$, $\mathfrak{p}[x_1, \cdots, x_n]$ is a prime ideal. Thus $f^{-1}(\mathfrak{p}) \subseteq V$ if and only if $I \subseteq \mathfrak{p}[x_1, \cdots, x_n]$ if and only if $I_A \subseteq \mathfrak{p}$ if and only if $\mathfrak{p} \in V(I_A)$. Then we get $\{\mathfrak{p} \in \operatorname{Spec} A \big| f^{-1}(\mathfrak{p}) \subseteq V\} = V(I_A)$ is a closed subset. Hence $f(\mathbb{A}^n_A \setminus V) = \operatorname{Spec} A \setminus V(I_A)$ is open subset. Conclude that f is an open map. $\square$

$\star$ Faithful Flatness

Proposition 6.22. Since the problem is local on the target, we may assume $Y = \operatorname{Spec} A$ for some ring A. Assume $\mathcal{F}$ and $\mathcal{G}$ are given by A-modules M and N respectively. Now suffices to show for all $y \in Y$, φ_y is an isomorphism.

As f is surjective, there is $x \in X$ such that $f(x) = y$. In addition, as f is flat, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$. Then by Proposition 3.19, $\mathcal{O}_{X,x}$ is faithfully flat over $\mathcal{O}_{Y,y}$. While $M_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to$

$N_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$ is isomorphic by assumption, we get $M_y \to N_y$ is isomorphic by Corollary 3.7. □

Corollary 6.6. Since S and T are affine over Y, we may assume $Y = \operatorname{Spec} A$, $S = \operatorname{Spec} B$ and $T = \operatorname{Spec} C$ for some $A, B \in \mathbf{Alg}_C$. Note that $\mathcal{O}_S$ and $\mathcal{O}_T$ can be identified with the quasi-coherent sheaves associated to B and C respectively. Then the morphism $S \to T$ corresponds to a morphism of A-algebras $C \to B$. By Proposition 6.22, we get $C \to B$ is an isomorphism and hence $S \to T$ is an isomorphism. □

8.6.5 Dimension Theory

Proposition 6.23. See in the website, dimension of irreducible affine variety is same as any open subset. □

Proposition 6.24. Since constructible can be written as finite union of locally closed subsets, assume that $Y = \cap_{i=1}^n U_i \cap V_i$. In particular, by taking irreducible components of each V_i, we may assume that V_i are irreducible. Then $\dim \overline{Y} \geq \dim Y \geq \max_i\{\dim(U_i \cap V_i)\}$. Note that $\overline{Y} = \cup_{i=1}^n \overline{U_i \cap V_i}$ and by Proposition 6.23, we immediately get

$$\dim \overline{Y} = \max_i\{\dim(\overline{U_i \cap V_i})\} = \max_i\{\dim V_i\} = \max_i\{\dim(U_i \cap V_i)\} \tag{42}$$

Thus $\dim Y = \dim \overline{Y}$. □

Proposition 6.25. By Proposition 3.22, we know for every open affine subset $U \subset X$, $U \times_k K$ is of pure dimension $\dim U = \dim X$. Note that $U \times_k K$ cover $X \times_k K$, for every irreducible component V of $X \times_k K$, $V \cap (U \times_k K) \neq \varnothing$ for some U and hence $V \cap (U \times_k K)$ is an irreducible component of $U \times_k K$ so that $\dim V = \dim(U \times_k K) = \dim X$. Conclude that X is of pure dimension $\dim X$. □

Corollary 6.7. Apply Proposition 6.25 to every irreducible component of X. □

Corollary 6.8. Since surjectivity is stable under base change, $X \times_k K \to X$ is surjective so that $\dim X \leq \dim(X \times_k K)$. Conversely, for each point $x \in X$, by Proposition 6.25 as K/k is finite, the fiber dimension $\dim(k(x) \times_k K) = \dim K = 0$. Hence $\dim(X \times_k K) \leq \dim X + 0 = \dim X$. Conclude that $\dim(X \times_k K) = \dim X$. □

8.6.6 Universal injectivity

Proposition 6.26. I don't know □

Proposition 6.27. As composition of projection map and $\Delta_{X/Y}$ is identity map of X, if f is universally injective, then projection map is injective and hence $\Delta_{X/Y}$ is surjective.

Conversely, for any morphism $Z \to Y$, it suffices to show that $\Delta_{X \times_Y Z/Z}$ is surjective. Note that there is a fibered diagram

$$\begin{tikzcd} X \times_Y Z \arrow[r, "\Delta_{X\times_Y Z/Z}"] \arrow[d] & X \times_Y Z \times_Z X \times_Y Z \cong X \times_Y X \times_Y Z \arrow[d] \\ X \arrow[r, "\Delta_{X/Y}"] & X \times_Y X \end{tikzcd} \tag{43}$$

Then as base change preserves surjectivity, we get $\Delta_{X\times_Y Z/Z}$ is surjective. $\square$

8.6.7 Vector bundles

Proposition 6.28. Assume $x_0,\ldots,x_n$ are the homogeneous coordinates of $\mathbb{P}^n_k$. Then transition functions of $\mathcal{O}_{\mathbb{P}^n_k}(-1)$ are given by $(D_+(x_i),\frac{x_i}{x_j})$. Define map $\mathcal{O}_{\mathbb{P}^n_k}(-1)\to\mathcal{O}^{\oplus n+1}_{\mathbb{P}^n_k}$ by sending s to $(\frac{x_0}{x_i}s,\ldots,\frac{x_n}{x_i}s)$ on $D_+(x_i)$. Then it is easy to check that these local maps glue together and give a global morphism $\mathcal{O}_{\mathbb{P}^n_k}(-1)\to\mathcal{O}^{\oplus n+1}_{\mathbb{P}^n_k}$. $\square$

Corollary 6.9. Take affine cover $\{U_i\}$ of X such that $\mathcal{E}\big|_{U_i}$ is trivial. Then we have a canonical inclusion $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(-1)\big|_{\pi^{-1}(U_i)}\to\pi^*\mathcal{E}\big|_{\pi^{-1}(U_i)}$ by Proposition 6.28. These local maps glue together and give a global morphism $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(-1)\to\pi^*\mathcal{E}$. $\square$

Proposition 6.29. "$\Rightarrow$ ": Assume $\mathbb{P}(\mathcal{E})$ is irreducible. Note that the natural morphism $\pi:\mathbb{P}(\mathcal{E})\to X$ is surjective. We know $X=\pi(\mathbb{P}(\mathcal{E}))$ is irreducible.

"$\Leftarrow$ ": Assume X is irreducible with generic point η. Then the fiber of π at η is a projective space over $k(\eta)$ and hence irreducible. Also, affine locally, $\mathbb{P}^N_A\to\operatorname{Spec}A$ is smooth so that π is smooth. By Proposition 6.20, we get $\mathbb{P}(\mathcal{E})$ is irreducible. $\square$

8.6.8 Exceptional locus

Proposition 6.30. Obviously, $\{x\in X\,|\,g \text{ is not defined at } f(x)\}\subseteq X\setminus U_X$. We only need to show the other containment. Assume g is defined at $f(x)$. Then by definition of rational map, there are open neighbourhoods $U\ni x$ and $V\ni x$ such that $f(U)\subseteq V$ and g is defined on V.

As U_X is open dense subset of X, $U_X\cap U$ is nonempty and dense in U. Hence $g\circ f\big|_U$ and $U\hookrightarrow X$ agree on $U_X\cap U$. By Proposition 6.8, $g\circ f\big|_U$ is just the open immersion.

Hence we can replace V by $V\cap g^{-1}(U)$, which is still an open neighbourhood of $f(x)$. Applying same argument for $f\circ g\big|_V$, we conclude that $U\subseteq U_X$ and $V\subseteq U_Y$, done! $\square$

Proposition 6.31. As D is not contained in $\operatorname{Sing}Y$, Y is regular at η so that $\mathcal{O}_{Y,\eta}$ is a DVR. Assume Z is an irreducible component of Y such that $D\subseteq Z$ and ξ is the generic point of Z, then $\mathcal{O}_{Y,\xi}=\operatorname{Frac}(\mathcal{O}_{Y,\eta})$.

Since U_Y is dense in Y, $\xi\in U_Y$ so that g induces a map $\operatorname{Spec}\mathcal{O}_{Y,\xi}\to U_X$. And we have a commutative diagram

$$\begin{tikzcd}
\operatorname{Spec}\mathcal{O}_{Y,\xi} \arrow[r] \arrow[d] & U_X \arrow[d, "f"] \\
\operatorname{Spec}\mathcal{O}_{Y,\eta} \arrow[r] \arrow[ur, dashed] & Y
\end{tikzcd} \tag{44}$$

Applying valuative criterion of properness, there exists unique lifting $\operatorname{Spec}\mathcal{O}_{Y,\eta}\to X$. Hence there exists $\delta\in U_X$ such that $f(\delta)=\eta$. By definition, we know $\eta\notin\operatorname{Ex}(g)$. $\square$

8.6.9 Ramification theory

Proposition 6.32. As f is a finite morphism of degree d, the stalk $(f_*\mathcal{O}_X)_Q$ is a free $\mathcal{O}_{Y,P}$-module of rank d. Tensoring by $k(Q)$, we get a d-dimensional vector space over $k(Q)$. Note that by Proposition 3.16, we have that

$$\begin{aligned}(f_*\mathcal{O}_X)_Q \otimes_{\mathcal{O}_{Y,Q}} k(Q) &\cong (\oplus_{i=1}^r \mathcal{O}_{X,P_i}) \otimes_{\mathcal{O}_{Y,Q}} k(Q) \\ &\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i}/\mathfrak{m}_Q\mathcal{O}_{X,P_i} \\ &\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i}/\mathfrak{m}_{P_i}^{e_{P_i}}\end{aligned} \tag{45}$$

Counting dimension, we get

$$d = \sum_{i=1}^r e_{P_i} \tag{46}$$

□

8.6.10 Intersection theory

Proposition 6.33. First note that $p^*(Z \cdot D^d) = p^*Z \cdot (p^*D)^d$ in Chow group. Suffice to show for all closed point $x \in X$ and $f^*[x] = \sum_i m_i[y_i]$, we have that $\chi(X, x) = \sum_i m_i \chi(X_K, y_i)$. In fact, this immediately comes from the equation

$$[k(x) : k] = \sum_i \mathrm{length}(\mathcal{O}_{X_K \times_X k(x), y_i}) \cdot [k(y_i) : K] \tag{47}$$

□

8.7 Homological Algebra 同调代数

Proposition 7.1. Here we only prove for the injective version. As $I^\bullet$ is acyclic, denote $\ker \partial_n = Z_n$. Then there are short exact sequences

$$0 \longrightarrow Z_n \longrightarrow I^n \longrightarrow Z_{n+1} \longrightarrow 0 \tag{48}$$

Since $I^\bullet$ is bounded-below, we may assume for all negative i, $I^i = 0$. Hence $Z_2 = I^1$. Consider the short exact sequence

$$0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow Z_3 \longrightarrow 0 \tag{49}$$

As I^1 is injective, the exact sequence splits so that Z_3 is a summand of I^2 and hence injective. By induction, we get that all those short exact sequences split. Now we have projections $p_n : I^n \to Z_n$ and inclusions $s_{n+1} : Z_{n+1} \to I^n$. Consider $D_n := s_n \circ p_n : I^n \to I^{n-1}$. In fact we have that

$$\partial \circ D_n + D_{n+1} \circ \partial = p_n + s_{n+1} = \mathrm{id}_{I^n} \tag{50}$$

Hence $I^\bullet$ is contractible. □